

The pair singular series of a polynomial II: zeros of Dedekind zeta functions in the second moment of prime values

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Abstract

[draft; numbers to be filled from WP0–WP3] Let $f \in \mathbb{Z}[t]$ be irreducible without fixed prime divisor, $K = \mathbb{Q}[t]/(f)$, and $S_f(h)$ its pair singular series. In part I we proved that the diagonal part of $\sum_{h \leq H} (S_f(h) - C(f)^2)$ has Dirichlet series $D_f(s) = \zeta_K(s+1)E_f(s)$. Here we determine the analytic structure of E_f completely: it is an infinite product of Artin L -functions of virtual representations of the Galois group, the first of which is $L(2s+2, \text{Sym}^2 V_f)^{-1} = (\zeta_K(2s+2) \prod_O \zeta_{K_O}(2s+2))^{-1}$ with K_O the fields cut out by the unordered pairs of roots; D_f is meromorphic in $\Re s > -1$, has real poles at $s = -1 + 1/k$, and $\Re s = -1$ is a natural boundary. Consequently the Riesz means of the diagonal satisfy an explicit formula, unconditional for $m \geq 2$, whose oscillating part is indexed by the zeros of these Dedekind zeta functions, with terms $x^{m-1+\rho/2}$, and whose smooth part contains $x^{m-2/3}$. For $f = t^2 + 1$ every coefficient is computed exactly and the formula is verified against the exact Riesz means to $x = 4 \cdot 10^7$: [WP0 result]. We prove $\Omega_{\pm}(x^{m-3/4})$ unconditionally, $O(x^{m-7/8+\varepsilon})$ after the real terms under GRH for ζ_K , and give the function-field version, where the formula is an exact finite sum. Assuming the Hardy–Littlewood pair conjecture with a power-saving error, these results describe the second-order term of the variance of the number of prime values of f ; we explain precisely why the zero-induced fluctuations lie below the scale at which that transfer is valid, so that no statement about prime values themselves and the zeros of ζ_K is claimed.

1 Introduction

Recollection of part I. [copy the definitions: $\omega_f, C(f), S_f, b(q), W_f, a_f(d) = W_f(d)\omega_f(d), A_f = a_f * 1$, Theorems 2–4 of part I.] The identity

$$\sum_{h \leq H} (S_f(h) - C^2) = -C^2 \sum_{d \geq 2} a_f(d) \left\{ \frac{H}{d} \right\} + C^2 \text{Off}_f(H)$$

splits the sum into a diagonal, generated by $D_f(s) = \sum_d a_f(d)d^{-s} = \zeta_K(s+1)E_f(s)$ with $\text{Res}_{s=0} D_f = 1/C(f)$, and an off-diagonal over pairs of distinct roots. Conjecture 1 of part I, $\sum_{h \leq H} (S_f(h) - C^2) = -\frac{1}{2}C \log H + A_f + o(1)$, is equivalent (Cesàro form) to $\text{Off}_f^*(H) = o(H \log H)$.

The linear case as template. For $f(t) = t$, Goldston and Suriajaya [3] proved that the Riesz means $\sum_{k \leq x} (x-k)^m \mathfrak{S}(k)$, $m \geq 2$, equal $x^{m+1}/(m+1) - \frac{1}{2}x^m(\log x - H_m + \gamma + \log 2\pi) + x^{m-1} \sum_{|\gamma| \leq U} a(\rho)x^{\rho/2} + O(x^{m-1+\varepsilon})$, the zeros of ζ entering through the factor $1/\zeta(2s+2)$ of the generating series; and $\Omega_{\pm}(x^{m-3/4})$. [cite also Vaughan 2001, Friedlander–Goldston 1995, GS JNT 2021.]

Results. [one paragraph per theorem, in the order of the sections.]

What is conditional on what. [Reuse part I's paragraph (A)/(B)/(C) and add:] (D) Everything in Sections 2–4 is unconditional and concerns the arithmetic function A_f ; the Riemann hypothesis for ζ_K enters only in the sharp form of the error term. (E) Section 7 is conditional on the Hardy–Littlewood pair conjecture for f with an error term $O(T^{1/2+\varepsilon})$, as in Montgomery–Soundararajan [6]; the zero terms of the explicit formula are of relative size $T^{-1/4}$, *below* that tolerance. Unconditionally, the low-lying zeros of ζ_K govern the second-order fluctuations of the Riesz means of S_f ; under a Bateman–Horn pair conjecture with power saving these means control the variance of prime values of f only at a scale above the zero-induced fluctuations; and under GRH the variance is in turn equivalent, by the Goldston–Montgomery theorem [2] in the form of Bui, Keating and Smith [1], to pair correlation of the zeros of ζ_K . We claim no unconditional link between prime values of f and zeros of ζ_K .

Relation to the literature. [from LITERATURE.md §0: Kuperberg–Rodgers–Roditty–Gershon (prime elements of K , $\zeta_K(s+1)$ in a singular-series average, no explicit formula); Bhowmik–Halupczok–Matsumoto–Suzuki and Goldston–Suriyaya 2023 (multi- L explicit formulas in the Goldbach setting); Hall–Keating–Roditty–Gershon and Hochfilzer (variances governed by higher-degree L -functions over $\mathbb{F}_q[t]$); no prior appearance of $\text{Sym}^2 V_f$ or of a polynomial singular-series explicit formula.]

2 The analytic structure of D_f

Throughout, $f \in \mathbb{Z}[t]$ is irreducible of degree n , without fixed prime divisor, $G = \text{Gal}(f)$, $V = V_f$ the permutation representation of G on the roots, $K = \mathbb{Q}[t]/(f)$, and for a prime p unramified in the splitting field, $g_p = \text{Frob}_p$ (a conjugacy class) with $\omega_f(p) = \chi_V(g_p)$ fixed points. We write $w = s + 1$, $u = u_p = p^{-1}$, $v = v_p = p^{-w}$. By part I, Theorem 3, $D_f(s) = \prod_p F_p(s)$ with

$$F_p(s) = P_p + p b(p) \omega_f(p) p^{-s} = \frac{1 - 2\omega_f(p)u + \omega_f(p)v}{(1 - \omega_f(p)u)^2} \quad (\omega_f(p) < p), \quad (1)$$

as one checks by writing $P_p = 1 - \omega^2 u^2 / (1 - \omega u)^2$ and $p b(p) \omega p^{-s} = \omega v / (1 - \omega u)^2$. The point of this section is that $F_p(s) \rightarrow 1 + \omega_f(p)v$ as $u \rightarrow 0$ with v fixed, and that $1 + \omega_f(p)v = 1 + \chi_V(g_p)v$ is *not* a finite product of Euler factors of Artin L -functions unless $\chi_V \leq 1$, i.e. unless f is linear.

2.1 The pair fields

Lemma 1 (pair fields). $\text{Sym}^2 V \cong V \oplus V^{(2)}$, where $V^{(2)}$ is the permutation representation of G on unordered pairs of distinct roots. Consequently

$$L(s, \text{Sym}^2 V) = \zeta_K(s) \prod_O \zeta_{K_O}(s),$$

the product over the G -orbits O on unordered pairs, K_O being the fixed field of the stabiliser of a pair in O . If G is 2-transitive there is one orbit, and $K^{(2)} = \mathbb{Q}(\alpha + \beta, \alpha\beta)$ has degree $n(n-1)/2$.

Proof. $\text{Sym}^2 V$ has the basis $e_\alpha e_\beta$ ($\alpha \leq \beta$ in any fixed order of the roots), which G permutes; the vectors e_α^2 span a copy of V and the vectors $e_\alpha e_\beta$, $\alpha \neq \beta$, span $V^{(2)}$. A permutation representation $\text{Ind}_H^G \mathbf{1}$ has Artin L -function ζ_{L^H} , L the splitting field; the stabiliser of the pair $\{\alpha, \beta\}$ fixes $\alpha + \beta$ and $\alpha\beta$, and $\mathbb{Q}(\alpha + \beta, \alpha\beta)$ has the right degree when G is 2-transitive. $\square$

For quadratics, $\text{Sym}^2 V = \mathbf{1} \oplus \mathbf{1} \oplus \chi_D$ and $L(s, \text{Sym}^2 V) = \zeta(s)^2 L(s, \chi_D)$; for an S_3 -cubic $K^{(2)} \cong K$ and $L(s, \text{Sym}^2 V) = \zeta_K(s)^2$; for an S_4 - or A_4 -quartic $K^{(2)}$ is the resolvent sextic field.

2.2 The plethystic exponents

Definition 2. For $N \geq 1$ and $g \in G$ put

$$\Psi_N(g) = \frac{1}{N} \sum_{j|N} \mu(j) (-1)^{N/j+1} \chi_V(g^j)^{N/j}.$$

Lemma 3. (i) Each Ψ_N is a virtual character of G , and $\Psi_1 = \chi_V$. (ii) For every $g \in G$, as formal power series in v ,

$$1 + \chi_V(g)v = \prod_{N \geq 1} \exp\left(\sum_{j \geq 1} \frac{\Psi_N(g^j)}{j} v^{Nj}\right) = \prod_{N \geq 1} \det(1 - gv^N | \Psi_N)^{-1},$$

where for a virtual character $\Psi = A - B$ we write $\det(1 - gX | \Psi)^{-1} = \det(1 - gX | A)^{-1} \det(1 - gX | B)$. (iii) $|\Psi_N(g)| \leq 2n^N/N$.

Proof. (ii) Taking logarithms, the claim is $\sum_{M \geq 1} c_M(g)v^M = \sum_{N,j} \Psi_N(g^j)v^{Nj}/j$ with $c_M(g) = (-1)^{M+1} \chi_V(g)^M/M$, i.e. $c_M(g) = \sum_{j|M} \Psi_{M/j}(g^j)/j$ for all M and g . Insert the definition: $\sum_{j|M} \frac{1}{j} \cdot \frac{j}{M} \sum_{i|M/j} \mu(i) (-1)^{M/(ij)+1} \chi_V(g^{ij})^{M/(ij)} = \frac{1}{M} \sum_{k|M} (-1)^{M/k+1} \chi_V(g^k)^{M/k} \sum_{i|k} \mu(i) = c_M(g)$, since only $k = 1$ survives. (i) Write $R(g, v) = 1 + \chi_V(g)v$; its coefficients lie in the representation ring $R(G)$. Suppose $\Psi_1, \dots, \Psi_{K-1}$ are virtual characters. The series $R(g, v) \prod_{N < K} \det(1 - gv^N | \Psi_N)$ has coefficients in $R(G)$ (products of the series $\det(1 - gX | A)^{\pm 1}$, whose coefficients are $\pm$ the characters of exterior or symmetric powers of A), and by (ii) it equals $1 + \Psi_K(g)v^K + O(v^{K+1})$; so $\Psi_K \in R(G)$. The case $N = 1$ is the definition. (iii) $|\chi_V(g^j)| \leq n$, and $\sum_{j|N} n^{N/j} \leq n^N + Nn^{N/2} \leq 2n^N$ for $n \geq 2$; for $n = 1$ the claim is trivial. $\square$

For f linear, $\chi_V = 1$ and $1 + v = (1 - v^2)/(1 - v)$: $\Psi_1 = 1$, $\Psi_2 = -1$, $\Psi_N = 0$ for $N \geq 3$, which is the factor $\zeta(s+1)/\zeta(2s+2)$ of the twin-prime case [3]. For $n \geq 2$ infinitely many Ψ_N are non-zero, because $1 + \chi_V(g)v$ with $\chi_V(g) \geq 2$ is not a finite product of cyclotomic polynomials in v .

2.3 The structure theorem

For a virtual character Ψ of G let $L(s, \Psi)$ be its Artin L -function (a quotient of two Artin L -functions of genuine representations, hence meromorphic in $\mathbb{C}$), with local factor $L_p(s, \Psi) = \det(1 - g_p p^{-s} | \Psi)^{-1}$ at unramified p ; and let S be the finite set of primes ramified in the splitting field or dividing the leading coefficient of f .

Theorem 4 (structure of D_f). For every $K \geq 1$,

$$D_f(s) = \prod_{N=1}^K L(N(s+1), \Psi_N) \cdot M_{f,K}(s), \quad M_{f,K}(s) = \prod_p F_p(s) \prod_{N=1}^K L_p(N(s+1), \Psi_N)^{-1}, \quad (2)$$

where the Euler product $M_{f,K}$ converges absolutely and locally uniformly in $\Re s > -1 + 1/(K+1)$, so that $M_{f,K}$ is holomorphic there. In particular $L(s+1, \Psi_1) = \zeta_K(s+1)$, so that $E_f = \prod_{N=2}^K L(Nw, \Psi_N) M_{f,K}$; D_f is meromorphic in $\Re s > -1$; and in $\Re s > -1 + 1/(K+1)$ its poles are among the poles of $\prod_{N \leq K} L(Nw, \Psi_N)$, while its zeros that are not zeros of that product are exactly the zeros of the local factors F_p , $p \notin S$, together with those of the finitely many local factors at $p \in S$.

Proof. Fix K and a compact set in $\Re w > 1/(K+1)$, on which $\Re w \geq \eta > 1/(K+1)$. For $p \notin S$ the local factor of $M_{f,K}$ is, by Lemma 3(ii),

$$M_p(s) = \frac{F_p(s)}{1 + \omega v} \cdot \frac{1 + \omega v}{\prod_{N \leq K} \det(1 - g_p v^N | \Psi_N)^{-1}} = \frac{1 - 2\omega u/(1 + \omega v)}{(1 - \omega u)^2} \cdot \exp\left(- \sum_{M > K} d_M(g_p) v^M\right),$$

with $\omega = \omega_f(p)$ and $d_M(g) = \sum_{N|M, N>K} \Psi_N(g^{M/N})N/M$, so $|d_M| \leq 2n^M \tau(M)$ by Lemma 3(iii). For p so large that $n|v| \leq np^{-\eta} \leq \frac{1}{2}$ and $\omega u \leq \frac{1}{4}$ we get $|\sum_{M>K} d_M v^M| \leq \sum_{M>K} 2\tau(M)(n|v|)^M \ll_K (n|v|)^{K+1} \ll p^{-(K+1)\eta}$, and $\frac{1-2\omega u/(1+\omega v)}{(1-\omega u)^2} = 1 + \frac{2\omega^2 uv}{1+\omega v} + O(u^2) = 1 + O(p^{-1-\eta}) + O(p^{-2})$. Since $(K+1)\eta > 1$, $\sum_p |M_p(s) - 1|$ converges uniformly on the compact set, which gives absolute and locally uniform convergence of $\prod_{p \notin S, p > p_0} M_p$; the finitely many remaining factors are rational functions of p^{-s} whose only possible poles are at $|p^{-Nw}| = 1$, i.e. on $\Re w = 0$. A uniformly convergent product of non-vanishing holomorphic factors is holomorphic and non-vanishing; hence $M_{f,K}$ vanishes exactly where a local factor does, and $M_p(s) = 0$ iff $F_p(s) = 0$ (the other factors of M_p are $\det(1 - g_p v^N | \Psi_N)$, non-zero for $\Re w > 0$). Letting $K \rightarrow \infty$ gives the continuation to $\Re s > -1$, the finitely many $L(Nw, \Psi_N)$ being meromorphic by Brauer's theorem. $\square$

Remark 5. $M_{f,K}$ is not $1 + O(p^{-2})$ at the level of individual primes: the mixed term $2\omega^2 uv/(1 + \omega v)$ is of size $p^{-1-\Re w}$. In the numerical work these terms are extracted as well ($[\zeta(w+1)L(w+1, \chi_D)]^4$, $[\zeta(2w+1)L(2w+1, \chi_D)]^{-8}$, ... for quadratics) so that the remaining product converges like $\sum p^{-2}$; see `explicit-diag.py`.

2.4 Quadratics: the exponents in closed form

For a quadratic f of discriminant D , $G = \{e, \tau\}$, $\chi_V = \mathbf{1} + \chi_D$, and every virtual character is $a\mathbf{1} + b\chi_D$ with $a, b \in \mathbb{Z}$. Write $\Psi_N = a_N\mathbf{1} + b_N\chi_D$, so that $L(Nw, \Psi_N) = \zeta(Nw)^{a_N} L(Nw, \chi_D)^{b_N}$, and let $M_N(x) = \frac{1}{N} \sum_{j|N} \mu(N/j) x^j$ be the necklace polynomial.

Proposition 6. *For every $N \geq 1$,*

$$a_N + b_N = \Psi_N(e) = -M_N(-2), \quad a_N - b_N = \Psi_N(\tau) = \frac{1}{N} \sum_{\substack{j|N \\ 2 \nmid j}} \mu(j) (-1)^{N/j+1} 2^{N/j}.$$

In particular $a_N = b_N = \frac{1}{2}M_N(2)$ for odd N , and for even N $a_N + b_N = -M_N(-2) < 0$; the first values are those of Table ??, and $a_1 = b_1 = 1$, i.e. $L(w, \Psi_1) = \zeta_K(w)$; $\Psi_2 = -\text{Sym}^2 V$; $a_3 = b_3 = 1$, so that $\zeta(3w)L(3w, \chi_D)$ appears in the numerator and D_f has a simple pole at $s = -\frac{2}{3}$; and $a_N > 0$ for all odd N , giving a pole of order $a_N = \frac{1}{2}M_N(2)$ at $s = -1 + 1/N$.

Proof. $\chi_V(e^j) = 2$ and $\chi_V(\tau^j) = 2$ or 0 according as j is even or odd; insert into Definition 2, using $(-1)^{k+1}2^k = -(-2)^k$ for the first formula. For odd N there is no even divisor, so $a_N = b_N = -\frac{1}{2}M_N(-2) = \frac{1}{2}M_N(2)$; $M_N(2) > 0$ counts binary Lyndon words. $\square$

2.5 The natural boundary

Theorem 7 (natural boundary). *Let f be irreducible of degree $n \geq 1$ without fixed prime divisor. Then D_f has infinitely many zeros in $-1 < \Re s < 0$, accumulating at every point of the line $\Re s = -1$, and $\Re s = -1$ is a natural boundary of D_f and of $\sum_n A_f(n)n^{-s} = \zeta(s)D_f(s)$. The same holds for $f(t) = t$, i.e. for the generating Dirichlet series of the twin-prime singular series.*

Proof. By (1), for $p \notin S$ with $\omega = \omega_f(p) \geq 1$, $F_p(s) = 0$ iff $\omega v = -(1 - 2\omega u)$, i.e. iff $p^{-w} = -(1 - 2\omega/p)/\omega$. These are the points

$$s_{p,j} = -1 + \frac{\log \omega - \log(1 - 2\omega/p)}{\log p} + \frac{(2j+1)\pi i}{\log p}, \quad j \in \mathbb{Z},$$

with $\Re s_{p,j} \rightarrow -1^+$ as $p \rightarrow \infty$ (for $\omega = 1$, $\Re s_{p,j} = -1 + 2/(p \log p) + O(p^{-2})$). By the Chebotarev density theorem there are infinitely many p with $\omega_f(p) \geq 1$ (indeed with $\omega_f(p) = n$). Given $t_0 \in \mathbb{R}$ and $\varepsilon > 0$, choose such a p with $\log p > \pi/\varepsilon$ and $|\Re s_{p,j} + 1| < \varepsilon$; the spacing of the

$\Im s_{p,j}$ is $2\pi/\log p < 2\varepsilon$, so some $s_{p,j}$ lies within 2ε of $-1 + it_0$. By Theorem 4 with K large, in a neighbourhood of $s_{p,j}$ we have $D_f = \Lambda_K \cdot M_{f,K}$ with $\Lambda_K = \prod_{N \leq K} L(Nw, \Psi_N)$ meromorphic and $M_{f,K}(s_{p,j}) = 0$; so $s_{p,j}$ is a zero of D_f unless it is a pole of Λ_K . The poles of Λ_K form a discrete set, so all but finitely many of the zeros $s_{p,j}$ in any bounded region are zeros of D_f . Now suppose D_f continued meromorphically to a disc B around $-1 + it_0$. Zeros of a meromorphic function accumulate only at poles, so $-1 + it_0$ would be a pole; but in a punctured neighbourhood of a pole $|D_f|$ is large, whereas zeros $s_{p,j}$ lie arbitrarily close to $-1 + it_0$: a contradiction. For $\zeta(s)D_f(s)$ the same argument applies, since $\zeta(s) \neq 0$ on $\Re s = -1$. For $f(t) = t$, $F_p = (1 - 2u + v)/(1 - u)^2$ vanishes at $p^{-w} = -(1 - 2/p)$ and the argument is identical. $\square$

Remark 8. For $f(t) = t$ the pure- v part of the local factor is $1 + v = (1 - v^2)/(1 - v)$, so the L -function part of D_f is just $\zeta(w)/\zeta(2w)$ and the contour in [3] can be moved to $\Re s = -1 + \varepsilon$, crossing the whole critical strip of $\zeta(2s + 2)$: this is why the linear case has an *unconditional* explicit formula with error $O(x^{m-1+\varepsilon})$, and why that error term cannot be improved by contour shifting. For $n \geq 2$ the factors $L(Nw, \Psi_N)$ with $N \geq 3$ pile up between $\Re s = -\frac{3}{4}$ and $\Re s = -1$, and the contour can only be moved a little past the first zero family.

3 The explicit formula for the diagonal

Let $R_m(x) = \sum_{n \leq x} (x - n)^m A_f(n)$, so that for $\Re s > 1$

$$R_m(x) = \frac{1}{2\pi i} \int_{(2)} G_m(s) x^{s+m} ds, \quad G_m(s) = \frac{m! \zeta(s) D_f(s)}{s(s+1) \cdots (s+m)}. \quad (3)$$

From now on f is quadratic with discriminant D , so that by Theorem 4 and Proposition 6, for every $K \geq 4$,

$$D_f(s) = \frac{\zeta(w)L(w, \chi_D) \zeta(3w)L(3w, \chi_D)}{\zeta(2w)^2 L(2w, \chi_D)} \prod_{N=4}^K \zeta(Nw)^{a_N} L(Nw, \chi_D)^{b_N} M_{f,K}(s), \quad w = s + 1. \quad (4)$$

3.1 Statement

Theorem 9. *Let f be an irreducible quadratic of discriminant D and $m \geq 2$. Assume the Riemann hypothesis for ζ and for $L(s, \chi_D)$. Then there is $\varepsilon_0 \in (0, \frac{1}{100})$ such that for $x \geq 2$ and some $U \in [x^{16}, 2x^{16}]$,*

$$R_m(x) = \frac{D_f(1)}{m+1} x^{m+1} + x^m \left(-\frac{\log x}{2C(f)} + c_{f,m}' \right) + c_{f,m}' x^{m-2/3} + x^{m-1} \sum_{\substack{\rho=\beta+i\gamma \\ |\gamma| \leq 2U}} Q_{m,\rho}(\log x) x^{\rho/2} + O(x^{m-3/4-\varepsilon_0}), \quad (5)$$

where ρ runs over the zeros of $\zeta(s)^2 L(s, \chi_D)$ (all with $\beta = \frac{1}{2}$), $Q_{m,\rho}$ is the polynomial (of degree less than the order of the pole of G_m at $\rho/2 - 1$, i.e. of degree 0 at simple zeros of L and 1 at simple zeros of ζ) such that $Q_{m,\rho}(\log x) x^{\rho/2-1+m}$ is the residue of $G_m(s) x^{s+m}$ at $s = \rho/2 - 1$, and

$$c_{f,m}' = \operatorname{Res}_{s=-2/3} G_m(s) = \frac{m! \zeta(-\frac{2}{3}) \zeta(\frac{1}{3}) L(\frac{1}{3}, \chi_D) L(1, \chi_D)}{3 \zeta(\frac{2}{3})^2 L(\frac{2}{3}, \chi_D)} \prod_{N \geq 4} \zeta(\frac{N}{3})^{a_N} L(\frac{N}{3}, \chi_D)^{b_N} \frac{M_{f,K}(-\frac{2}{3})}{(-\frac{2}{3})(\frac{1}{3}) \cdots (m - \frac{2}{3})},$$

$c_{f,m}'$ being the constant term of the residue at $s = 0$. The implied constant depends on f , m and ε_0 .

The coefficient of $x^m \log x$ is $-1/(2C(f))$ because $\operatorname{Res}_{s=0} D_f = 1/C(f)$ (part I, Theorem 3) and $\zeta(0) = -\frac{1}{2}$; explicitly $c_{f,m}' = -\frac{1}{2C(f)} (\log 2\pi - H_m) - \frac{1}{2} \kappa_f$ with κ_f the constant term of the Laurent expansion of D_f at 0 and $H_m = \sum_{k \leq m} 1/k$.

3.2 Growth lemmas

We use the following consequences of the Riemann hypothesis for ζ and $L(s, \chi_D)$ ([8, Chapter 14] for ζ ; the proofs transfer verbatim to $L(s, \chi_D)$, see [4, §5.7]). Let $\Lambda \in \{\zeta, L(\cdot, \chi_D)\}$ and $t \geq 2$.

(G1) For every fixed $\sigma > \frac{1}{2}$: $\Lambda(\sigma + it)^{\pm 1} \ll t^\varepsilon$; for every fixed σ : $\Lambda(\sigma + it) \ll t^{\max(0, 1/2 - \sigma) + \varepsilon}$, and for fixed $\sigma < \frac{1}{2}$: $\Lambda(\sigma + it)^{-1} \ll t^\varepsilon$ (by the functional equation, $|\Lambda(\sigma + it)| \asymp t^{1/2 - \sigma} |\Lambda(1 - \sigma - it)|$).

(G2) Unconditionally, for $-1 \leq \sigma \leq 2$: $\frac{\Lambda'}{\Lambda}(\sigma + it) = \sum_{|\gamma - t| \leq 1} \frac{1}{\sigma + it - \rho} + O(\log t)$, the sum over zeros $\rho = \beta + i\gamma$ of Λ , and the number of zeros with $|\gamma - t| \leq 1$ is $\leq c_0 \log t$.

Lemma 10 (horizontal segments). *Assume RH for ζ and $L(s, \chi_D)$. For every $\delta > 0$ there is $\varepsilon_1(\delta) > 0$ such that for all $T \geq T_0(\delta)$ and $\varepsilon_0 \leq \varepsilon_1$, there exists $U \in [T, 2T]$ with*

$$|\zeta(\sigma + 2iU)^{-1}| + |L(\sigma + 2iU, \chi_D)^{-1}| \leq T^\delta \quad \text{uniformly for } \frac{1}{2} - 2\varepsilon_0 \leq \sigma \leq 2.$$

Proof. Let Λ be one of the two functions, $\sigma_1 = \frac{1}{2} + \varepsilon_0$, and $Z = [\frac{1}{2} - 2\varepsilon_0, \sigma_1]$ (the “zone”). For $\sigma \geq \sigma_1$ the claim is (G1) with $\delta/2$ in place of ε . For $\sigma \in Z$ and $t = 2U$ write, by (G2),

$$\log \frac{1}{|\Lambda(\sigma + it)|} = \log \frac{1}{|\Lambda(\sigma_1 + it)|} + \Re \int_{\sigma}^{\sigma_1} \frac{\Lambda'}{\Lambda}(\sigma' + it) d\sigma' \leq \frac{\delta}{4} \log T + \sum_{|\gamma - t| \leq 1} \int_Z \frac{d\sigma'}{|\sigma' + it - \rho|} + 3\varepsilon_0 C \log t.$$

Under RH $\beta = \frac{1}{2}$, so $|\sigma' + it - \rho| = \sqrt{(\sigma' - \frac{1}{2})^2 + (t - \gamma)^2}$ and, with $\theta = |t - \gamma|$, $\int_Z \leq \int_{-2\varepsilon_0}^{2\varepsilon_0} \frac{dy}{\sqrt{y^2 + \theta^2}} = 2 \operatorname{arsinh}(2\varepsilon_0/\theta)$, which is $\leq 4\varepsilon_0/\theta$ for $\theta \geq 2\varepsilon_0$ and $\leq 2 \log(4\varepsilon_0/\theta) + 2$ for $\theta < 2\varepsilon_0$.

By (G2) the zeros with $2\varepsilon_0 \leq \theta \leq 1$ contribute at most $4\varepsilon_0 \sum \theta^{-1} \leq 4\varepsilon_0 \log(1/2\varepsilon_0) c_0 \log t$ (split the range of θ dyadically: each dyadic block $[\theta_0, 2\theta_0]$ contains at most $c_0 \log t$ zeros, each contributing at most θ_0^{-1}). The zeros with $\theta < 2\varepsilon_0$ contribute $2\Sigma_\Lambda(t) + 2\#\{\gamma : |\gamma - t| < 2\varepsilon_0\}$ with $\Sigma_\Lambda(t) = \sum_{|\gamma - t| < 2\varepsilon_0} \log(4\varepsilon_0/|t - \gamma|)$. Now average over $t \in [2T, 4T]$: each zero contributes at most $\int_{-2\varepsilon_0}^{2\varepsilon_0} \log(4\varepsilon_0/|y|) dy = 4\varepsilon_0(1 + \log 2)$, and there are $\leq c'_0 T \log T$ zeros with $\gamma \in [2T - 1, 4T + 1]$, so $\frac{1}{2T} \int_{2T}^{4T} \Sigma_\Lambda(t) dt \leq 6c'_0 \varepsilon_0 \log T$ and similarly the average of $\#\{\gamma : |\gamma - t| < 2\varepsilon_0\}$ is $\leq 4c'_0 \varepsilon_0 \log T$. By Markov's inequality the set of $t \in [2T, 4T]$ with $\Sigma_\zeta(t) + \Sigma_L(t) + \#\zeta + \#L \leq 100c'_0 \varepsilon_0 \log T$ has measure $\geq \frac{3}{5} \cdot 2T$; pick U with $2U$ in it. Altogether $\log(1/|\Lambda(\sigma + 2iU)|) \leq (\frac{\delta}{4} + c_1 \varepsilon_0 \log(1/\varepsilon_0)) \log T$ for $\sigma \in Z$, with c_1 absolute, which is $\leq \delta \log T$ once $\varepsilon_0 \leq \varepsilon_1(\delta)$. $\square$

3.3 Proof of Theorem 9

Fix $K = 15$ (any $K \geq 4$ with $(K + 1)(\frac{1}{4} - \varepsilon_0) > 1$ works), $\delta = \varepsilon_0$ and $\varepsilon_0 \leq \min(\varepsilon_1(\varepsilon_0), \frac{1}{100})$, which is possible since $\varepsilon_1(\delta) \gg \delta/\log(1/\delta)$. Let U be as in Lemma 10 with $T = x^{16}$, and let $\mathcal{R}$ be the rectangle with vertices $2 \pm iU$ and $-\frac{3}{4} - \varepsilon_0 \pm iU$. By (3) (truncating the integral at height U costs $\ll x^{m+2}U^{-m}$, since $G_m(2 + it) \ll t^{-m-1}$) and the residue theorem,

$$R_m(x) = \sum_{\text{poles in } \mathcal{R}} \operatorname{Res}(G_m(s)x^{s+m}) + \frac{1}{2\pi i} \left(\int_{-3/4 - \varepsilon_0 - iU}^{-3/4 - \varepsilon_0 + iU} + \int_{\text{horiz.}} \right) G_m(s)x^{s+m} ds + O(x^{m+2}U^{-m}).$$

Poles. By (4) and Theorem 4, in $\Re s \geq -\frac{3}{4} - \varepsilon_0 > -\frac{4}{5}$ the poles of G_m are: $s = 1$ (simple, from $\zeta(s)$; $D_f(1) = \sum_d a_f(d)/d$), $s = 0$ (double: the pole of $\zeta(w)$ and the factor $1/s$), $s = -\frac{2}{3}$ (simple, from $\zeta(3w)$), and $s = \rho/2 - 1$ for the zeros ρ of $\zeta(2w)^2 L(2w, \chi_D)$ with $|\gamma| \leq 2U$; the factors $\zeta(Nw)^{a_N} L(Nw, \chi_D)^{b_N}$, $N \geq 4$, have $\Re(Nw) \geq 1 - 4\varepsilon_0 > \frac{1}{2}$ there, so under RH they have neither zeros nor poles except $\zeta(Nw)$ at $Nw = 1$, i.e. $s = -1 + 1/N \leq -\frac{3}{4}$, outside $\mathcal{R}$ for $N \geq 4$ (and $N = 4$ gives a zero of D_f); $M_{f,K}$ is holomorphic in $\Re s > -1 + \frac{1}{16}$; and $s = -1, \dots, -m$

are outside $\mathcal{R}$. The residues at $1, 0, -\frac{2}{3}$ are the three explicit terms of (5), and the residue at $\rho/2 - 1$ is $Q_{m,\rho}(\log x)x^{\rho/2-1+m}$ by definition.

Vertical line. On $\Re s = -\frac{3}{4} - \varepsilon_0$ (so $\Re w = \frac{1}{4} - \varepsilon_0$, $\Re 2w = \frac{1}{2} - 2\varepsilon_0$, $\Re 3w = \frac{3}{4} - 3\varepsilon_0$, $\Re Nw \geq 1 - 4\varepsilon_0$ for $N \geq 4$), (G1) gives, for $|t| \geq 2$, $\zeta(s) \ll |t|^{5/4+\varepsilon_0+\varepsilon}$, $\zeta(w)L(w, \chi_D) \ll |t|^{1/2+2\varepsilon_0+\varepsilon}$, $\zeta(2w)^{-2}L(2w, \chi_D)^{-1} \ll |t|^\varepsilon$, $\zeta(3w)L(3w, \chi_D) \ll |t|^\varepsilon$, $\prod_{N \geq 4} \ll |t|^\varepsilon$, and $M_{f,K} \ll 1$; hence $G_m(s) \ll |t|^{7/4+3\varepsilon_0+\varepsilon-(m+1)}$, and for $m \geq 2$ the exponent is < -1 . The vertical integral is therefore $\ll x^{m-3/4-\varepsilon_0}$.

Horizontal segments. On $\Im s = \pm U$, $-\frac{3}{4} - \varepsilon_0 \leq \sigma \leq 2$: (G1) bounds $\zeta(s)$, $\zeta(w)L(w)$, $\zeta(3w)L(3w)$ and the $N \geq 4$ factors by $U^{7/4+3\varepsilon_0+\varepsilon}$ (worst case at the left end), Lemma 10 bounds $\zeta(2w)^{-2}L(2w, \chi_D)^{-1}$ by $U^{3\varepsilon_0}$, and $M_{f,K} \ll 1$; the kernel is $\ll U^{-m-1}$ and $|x^{s+m}| \leq x^{m+2}$. The two segments contribute $\ll x^{m+2}U^{7/4+6\varepsilon_0+\varepsilon-m-1} \leq x^{m+2}U^{-1/4+7\varepsilon_0}$ for $m \geq 2$, and the truncation of the original integral at height U contributes $\ll x^{m+2}U^{-m}$; with $U \geq x^{16}$ and $\varepsilon_0 \leq \frac{1}{100}$ both are $\ll x^{m-1}$.

Collecting, $R_m(x)$ equals the residue sum plus $O(x^{m-3/4-\varepsilon_0})$, which is (5). $\square$

Remark 11 (on the zero sum). The sum over $|\gamma| \leq 2U$ is not absolutely convergent as $U \rightarrow \infty$ with the polynomial growth of the coefficients; as in [3, Theorem 3], simplicity of the zeros and a bound $\sum_{|\gamma| \leq T} |\zeta'(\rho)|^{-2} \ll T$ (Gonek–Hejhal) would make $x^{m-3/4} \sum_{\gamma} Q_{m,\rho} x^{i\gamma/2}$ absolutely convergent for $m \geq 2$. Numerically the coefficients decay like $|\gamma|^{-5/4}$ for $m = 2$.

3.4 $f = t^2 + 1$: every coefficient

For $f = t^2 + 1$, $D = -4$, $\chi_D = \chi_{-4}$, $K = \mathbb{Q}(i)$, and every quantity in (5) is computable: $D_f(1) = 0.7507908137$, $C(f) = 1.372813462821$ (from $L(1, \chi_{-4}) = \pi/4$ and $M_{f,K}(0)$; part I quotes $1.3728134628\dots$), $c_{f,2} = -0.454974899864$, $c_{f,3} = -0.333569719058$, and the coefficients $Q_{m,\rho}$ for the zeros with $\gamma \leq 100$ are computed as Taylor coefficients of $(s - \rho/2 + 1)^{\text{ord}} G_m(s)$ by trapezoidal Cauchy integrals on circles of radius 0.015 (`explicit-diag.py`; $M_{f,K}$ with $K = 15$ and primes up to $4 \cdot 10^6$, with the tails of the mixed terms handled by the prime number theorem). [Table of $Q_{2,\rho}$ for the first 8 zeros of $L(\chi_{-4})$ and 6 of ζ ; generate from the cached residues.]

3.5 Numerical verification

$f = t^2 + 1$, $m = 2$: data minus exact smooth terms (grey) and predicted zero sum (blue)

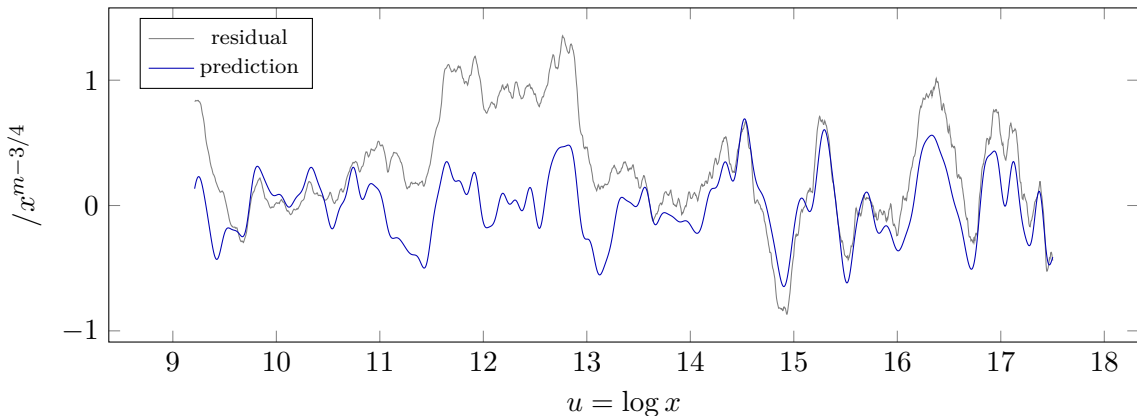


Figure 1: The residual of the diagonal Riesz mean of order 2 of $t^2 + 1$ after subtracting the exact terms at $s = 1$, $s = 0$ and the real poles $s = -2/3, -4/5, -6/7$, against the sum over the zeros of $L(s, \chi_{-4})$ and of ζ with $\gamma \leq 100$. Regression coefficient $\beta = 1.008$ (0.991 with a smooth nuisance basis); per family $\beta_\zeta = 1.006$, $\beta_L = 1.020$.

The raw Riesz means $R_m(x)$, $m = 1, 2, 3$, were computed exactly (compensated summation,

$A_f(n)$ from $\omega_f(p) = 1 + \chi_{-4}(p)$) at 4096 points equally spaced in $u = \log x \in [\log 10^4, \log 4 \cdot 10^7]$. From R_2 we subtract the exact terms at $s = 1$, $s = 0$, $-\frac{2}{3}$, $-\frac{4}{5}$ (order 3) and $-\frac{6}{7}$ (order 9); the only fitted quantity is a correction to the x^{m+1} coefficient, which comes out at $-1.0 \cdot 10^{-12}$ relative to $D_f(1)/(m+1)$ and is itself a check of $D_f(1)$ to twelve digits. The residual, divided by $x^{5/4}$, has root mean square 0.528. The predicted zero sum (zeros of $L(s, \chi_{-4})$ and of ζ with $\gamma \leq 100$) has root mean square 0.27. Regressing the residual on the prediction gives $\beta = 1.008$ (Figure 1); with a smooth nuisance basis $x^{m-8/9}(\log x)^j$, $j \leq 3$, absorbing the dropped real poles, $\beta = 0.991$ with $R^2 = 0.55$ and a remaining root mean square of 0.291; the two families separately give $\beta_\zeta = 1.006$ and $\beta_L = 1.020$. Controls in which all ordinates are shifted by $\delta \in [-0.6, 0.6]$ give β between -0.94 and 0.68 and negative R^2 . For $m = 3$ the zero terms (rms 0.11) lie below the smooth leftover (rms 0.6); with the nuisance basis $\beta = 0.998$. The remaining rms 0.29 is accounted for by the truncation at $\gamma \leq 100$ (the coefficients decay only like $\gamma^{-5/4}$) and the $O(x^{m-3/4-\varepsilon_0})$ term. We regard this as a confirmation of Theorem 9 with the predicted amplitudes and phases to about 1%; it does *not* resolve individual ordinates closer than $4\pi/8.3 \approx 1.5$. [say what happened when the contour was pushed to $-7/8$: the second family has rms 13 and 92 and $\beta = 0.001$.]

4 Ω -results and conditional bounds

Theorem 12. *Let f be an irreducible quadratic and $m \geq 1$, and let $E_m(x) = R_m(x) - \frac{D_f(1)}{m+1}x^{m+1} - x^m(-\frac{\log x}{2C(f)} + c_{f,m}) - c'_{f,m}x^{m-2/3}$. Suppose that for some zero $\rho_1 = \frac{1}{2} + i\gamma_1$ of $\zeta(s)^2L(s, \chi_D)$ the function G_m has a genuine pole at $\rho_1/2 - 1$ (i.e. $Q_{m,\rho_1} \neq 0$). Then, unconditionally,*

$$E_m(x) = \Omega_\pm(x^{m-3/4}).$$

For $f = t^2 + 1$ the hypothesis holds with $\gamma_1 = 6.0209\dots$, the first zero of $L(s, \chi_{-4})$ ($|Q_{2,\rho_1}|$ is computed in Section 3.4 and is non-zero).

Proof. Suppose $E_m(x) \leq \varepsilon x^{m-3/4}$ for all $x \geq x_0$. The function $F(x) = \varepsilon x^{m-3/4} - E_m(x)$ is then non-negative for $x \geq x_0$, and for $\Re s > 1$

$$\int_{x_0}^{\infty} F(x) x^{-s-m-1} dx = \frac{\varepsilon x_0^{-s-1/4}}{s + \frac{1}{4}} - G_m(s) + \left(\text{the Mellin transforms of the subtracted terms} \right) + (\text{entire}),$$

where we used $\int_1^{\infty} R_m(x) x^{-s-m-1} dx = G_m(s)$ for $\Re s > 1$ (Mellin inversion of (3)) and the fact that $\int_1^{x_0}$ is entire; the Mellin transforms of x^{m+1} , $x^m \log x$, x^m , $x^{m-2/3}$ are $1/(s-1)$, $1/s^2$, $1/s$, $1/(s+\frac{2}{3})$ up to constants, and they cancel the poles of G_m at 1 , 0 , $-\frac{2}{3}$ exactly by the definition of the coefficients. The right-hand side is therefore meromorphic in $\Re s > -\frac{4}{5}$ (Theorem 4), holomorphic on the real segment $(-\frac{4}{5}, \infty)$ (the remaining real singularities of D_f in this range are the zeros at $-\frac{1}{2}$ and $-\frac{3}{4}$), and has a pole at $\rho_1/2 - 1$, of real part $-\frac{3}{4}$. By Landau's theorem for integrals with non-negative integrand [7, Theorem 1.7], the abscissa of convergence of the left-hand side is a singularity of the function on the real axis; hence the integral converges for $\Re s > -\frac{4}{5}$, and its value is holomorphic there, contradicting the pole at $\rho_1/2 - 1$. Thus $E_m(x) > \varepsilon x^{m-3/4}$ for arbitrarily large x , i.e. $E_m = \Omega_+(x^{m-3/4})$; applying the argument to $-E_m$ gives Ω_- . $\square$

Remark 13. (i) No positivity of A_f is needed, and no hypothesis beyond the existence of one uncanceled pole on $\Re s = -\frac{3}{4}$; if $\zeta^2 L(\chi_D)$ had a zero with $\beta > \frac{1}{2}$ the same argument gives $\Omega_\pm(x^{m-1+\beta/2})$, provided that pole is uncanceled. (ii) Under RH for ζ and $L(s, \chi_D)$, Theorem 9 gives $E_m(x) \ll x^{m-3/4+\varepsilon}$ for $m \geq 2$ provided the zero sum is $O(x^\varepsilon)$ in the normalisation $x^{m-3/4}$, which follows from the simplicity and Gonek–Hejhal hypotheses of the remark above; so under those hypotheses the exponent $m - \frac{3}{4}$ is exact. (iii) For $f = t^2 + t + 41$ the first zero

of $L(s, \chi_{-163})$ has $\gamma_1 = 0.2029$, so the Ω -oscillation has period $4\pi/\gamma_1 \approx 62$ in $\log x$: over any computable range the term looks like a constant multiple of $x^{m-3/4}$. The family $\{L(s, \chi_D)\}$ has symplectic symmetry [5], so such a low zero is the class-number-one exception, not the rule.

5 The off-diagonal

[Route A of routes.tex: Prop. A1 (proved), A2/A3 (to prove) for quadratics. New here: the off-diagonal Dirichlet series $\sum_d d^{-s} W_f(d) \sum_{s \neq s'} \zeta(s, \langle (s' - s)/d \rangle)$ and the statement that Conjecture 1 of part I, in analytic form, is its continuation to a pole-free strip left of 0. Numerical: $\text{Off}_f(H)$ for $t^2 + 1$ to 10^7 and its spectrum. If nothing more: state clearly that the explicit formula of Section 3 is for the diagonal and that the full sum adds an $O(H)$ -in-Cesàro-form term whose fluctuations are those of the fractional parts of the root differences.]

6 The function field $\mathbb{F}_q[u]$

[Route B. New: with $T = q^{-s}$ the diagonal generating function is a rational function of T (Prop. B2 of routes.tex + Theorem 4), so the Riesz-mean analogue is an EXACT finite sum over the inverse roots: the Weil zeros of $L(T, \chi_D)$ and of the extracted factors, and the real "poles" $T = q^{1/k}$. Write it down for $f = t^2 - D$; compare with thesis/ff.ts data. Then Prop. B3, $q \rightarrow \infty$, for the off-diagonal (Katz/Deligne).]

7 Consequences for prime values

[Conditional section. (i) Under the HL pair conjecture with error $O(T^{1/2+\varepsilon})$, the variance formula of part I with the diagonal explicit formula inserted: the polynomial Montgomery–Soundararajan law plus an explicit oscillatory correction of relative size $T^{-1/4}$, which is below the error tolerated by the hypothesis — say so. (ii) Goldston–Montgomery for ζ_K : what Bui–Keating–Smith give for the degree-2 non-primitive $\zeta_K = \zeta L(\chi_D)$ (union of two independent GUE processes, Murty–Perelli), and why no analogue exists for prime values of a non-linear f (no Dirichlet series). (iii) What GRH for ζ_K does say as a theorem: Grenié–Molteni–Perelli for prime ideals; our $O(x^{m-7/8+\varepsilon})$.]

8 Discussion and open problems

[(1) Prove Theorem ?? in general. (2) The off-diagonal for quadratics (Route A). (3) Is the natural boundary known for the Goldston–Suriyaya series? (4) Higher moments: R_k for polynomial tuples and Gaussian fluctuations. (5) The constant A_f in closed form.]

Data and code

followup/scripts/riesz-raw.ts (raw Riesz means), followup/scripts/explicit-diag.py (exponents, constants, residues, comparison), followup/data/*. [add WP2 scripts]

Statement on authorship and the use of AI

[as in part I]

References

- [1] H. M. Bui, J. P. Keating, and D. J. Smith. On the variance of sums of arithmetic functions over primes in short intervals and pair correlation for L -functions in the Selberg class. *J. London Math. Soc. (2)*, 94(1):161–185, 2016. arXiv:1506.03741.

- [2] D. A. Goldston and H. L. Montgomery. Pair correlation of zeros and primes in short intervals. In *Analytic Number Theory and Diophantine Problems (Stillwater, OK, 1984)*, volume 70 of *Progress in Mathematics*, pages 183–203. Birkhäuser, 1987.
- [3] D. A. Goldston and A. I. Suriajaya. A singular series average and the zeros of the Riemann zeta-function. *Acta Arithmetica*, 2021. arXiv:2007.16099.
- [4] Henryk Iwaniec and Emmanuel Kowalski. *Analytic Number Theory*, volume 53 of *American Mathematical Society Colloquium Publications*. American Mathematical Society, 2004.
- [5] Nicholas M. Katz and Peter Sarnak. Zeroes of zeta functions and symmetry. *Bull. Amer. Math. Soc. (N.S.)*, 36(1):1–26, 1999.
- [6] H. L. Montgomery and K. Soundararajan. Primes in short intervals. *Communications in Mathematical Physics*, 252:589–617, 2004.
- [7] Hugh L. Montgomery and Robert C. Vaughan. *Multiplicative Number Theory I. Classical Theory*, volume 97 of *Cambridge Studies in Advanced Mathematics*. Cambridge University Press, 2007.
- [8] E. C. Titchmarsh. *The Theory of the Riemann Zeta-Function*. Oxford University Press, second edition, 1986. Revised by D. R. Heath-Brown.